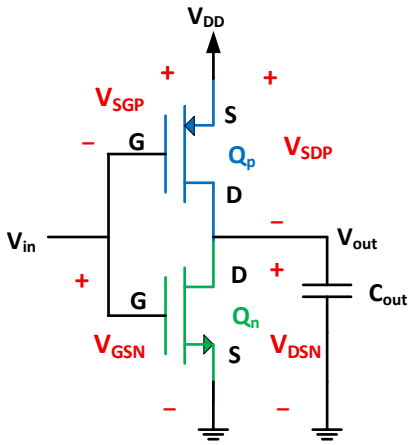
ECE321 Equation Sheet 5: Digital Logic

Transistor liner triode resistance:

$$R_{DSN} = \frac{1}{\mu_n C_{ox} (W/L)_n (V_{GSN} - V_{tn})} \qquad R_{DSP} = \frac{1}{\mu_p C_{ox} (W/L)_p (V_{SGP} - |V_{tp}|)}$$

NMOS: PDN $\rightarrow$ 0V PUN $\rightarrow$ $V_{DD} - V_{tn}$ **PMOS**: PDN $\rightarrow |V_{tp}|$ PUN $\rightarrow$ V_{DD}

CMOS Inverter



PUN: All PMOS $\rightarrow Y$ **PDN**: All NMOS $\rightarrow \overline{Y}$

DeMorgan's Law: $\overline{A \cdot B} = \overline{A} + \overline{B}$ $\overline{A + B} = \overline{A} \cdot \overline{B}$

(1) Complement all variables; (2) swap AND/OR inside out; (3) Complement entire expression

Switches in series: AND Switches in parallel: OR PUN: Must complement inputs

Synthesis: PUN $\rightarrow Y$ – DeMorgan's Law if complement over entire expression;

complement input variables, series -> AND, parallel -> OR

PDN $\rightarrow \overline{Y}$ -- DeMorgan's Law if complement over entire expression;

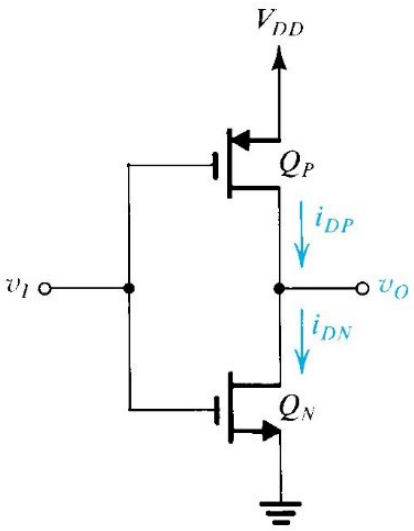
No complement of input variables, series -> AND, parallel -> OR

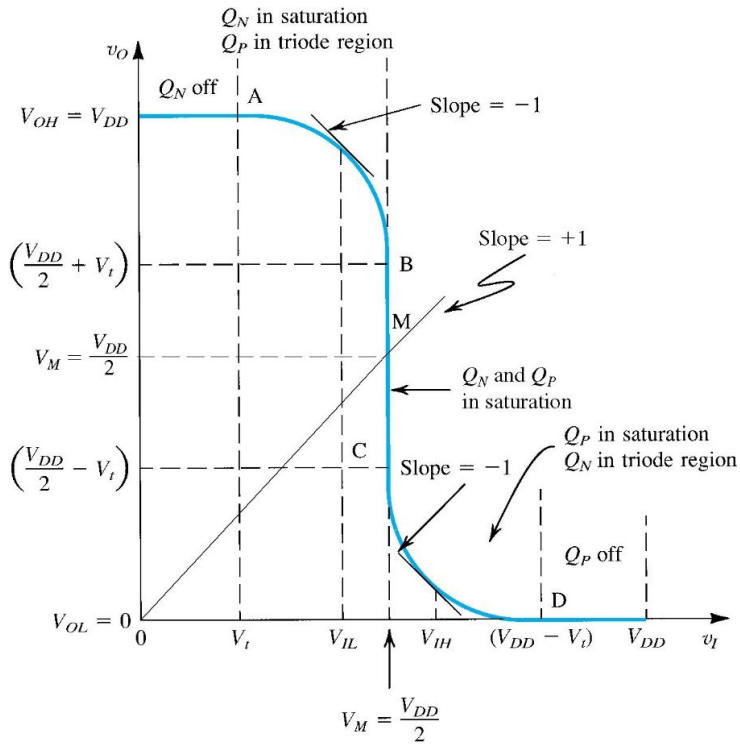
Noise Margins

$$NML = V_{IL} - V_{OL} \qquad NMH = V_{OH} - V_{IH}$$

Matched Inverter: (1) $V_{tn} = |V_{tp}|$ (2) $K_n = K_p \rightarrow \mu_n \times W_n = \mu_p \times W_P$

CMOS Inverter VTC:





$$V_{DSN} = V_{OUT} \qquad V_{GSN} = V_{IN} \qquad V_{SDP} = V_{DD} - V_{OUT} \qquad V_{SGP} = V_{DD} - V_{IN}$$

NMOS: *CUT:* $V_{GSN} < V_t \rightarrow V_{IN} < V_t$ *SAT*: $V_{DSN} > V_{GSN} - V_t \rightarrow V_{OUT} > V_{IN} - V_t$

PMOS: *CUT:* $V_{SGP} < V_t \rightarrow V_{IN} > V_{DD} - V_t$ *SAT*: $V_{SDP} > V_{SGP} - V_t \rightarrow V_{OUT} < V_{IN} + V_t$

V_{IH} Derivation: $I_{DN}(\text{TRI}) = I_{DP}(\text{SAT})$ $V_{IN} = V_{IH}$ $\dfrac{dV_o}{dV_{IN}} = -1$

$$V_o\left(V_{IN} = V_{IH}\right) = V_{IH} - \frac{1}{2}V_{DD} \qquad V_{IH} = \frac{1}{8}\left(5V_{DD} - 2V_t\right)$$

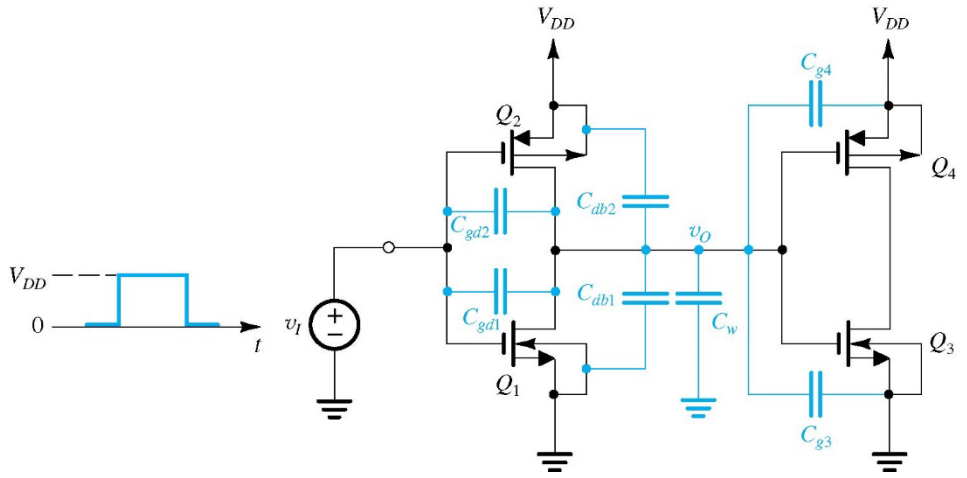
$$V_{IL} = \frac{1}{8}\left(3V_{DD} + 2V_t\right) \qquad \text{NMH} = \text{NML} = \frac{1}{8}\left(3V_{DD} + 2V_t\right)$$

CMOS Inverter VTC Asymmetry: $I_{DN}(\text{SAT}) = I_{DP}(\text{SAT})$ $V_{IN} = V_M$

$$r = \sqrt{\frac{K_p}{K_n}} = \sqrt{\frac{\mu_p \times W_p}{\mu_n \times W_n}} \qquad V_M = \frac{V_{tn} + r\left(V_{DD} - \left|V_{tp}\right|\right)}{1 + r}$$

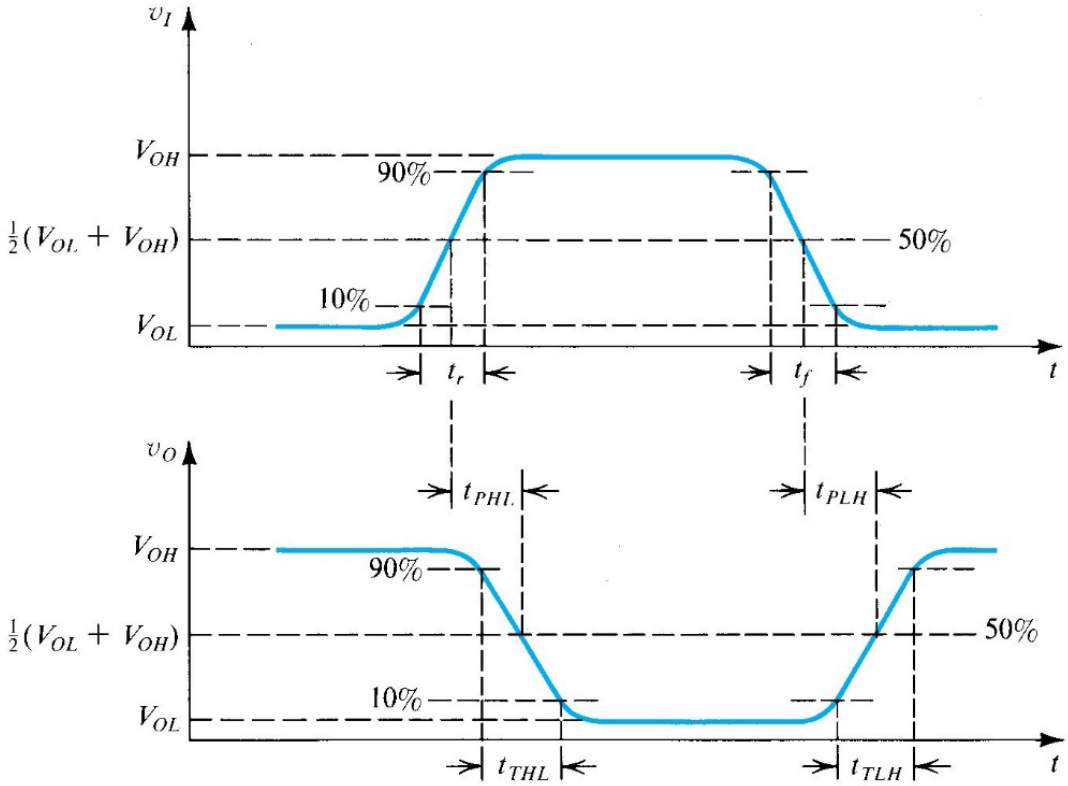
$$\Delta V_M = V_M - \frac{V_{DD}}{2} \qquad NML = NML_{matched} + \Delta V_M \qquad NMH = NMH_{matched} - \Delta V_M$$

Silicon Area: $A = L_p \times W_p + L_n \times W_n$



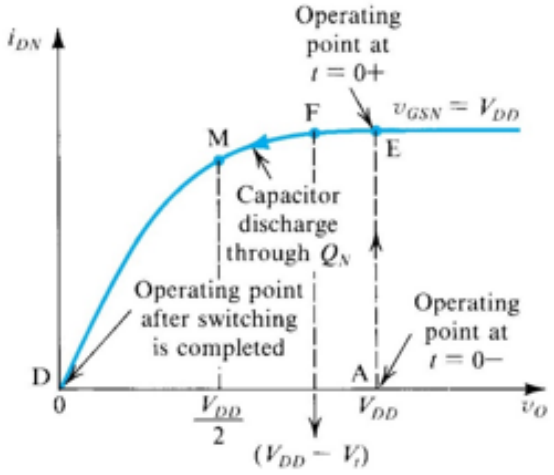
CMOS Inverter Dynamic Operation

Definitions: rise time, fall time, propagation delay times, transition times



Propagation Delay Calculations

Method 1: *Average Current*



$$I_{DN,ave} = \frac{I_{DN}(E)+I_{DN}(M)}{2} \qquad i_C = C\frac{dV_C}{dt}$$

$$a_n = \frac{2}{\frac{7}{4} - 3\left(\frac{V_{tn}}{V_{DD}}\right) + \left(\frac{V_{tn}}{V_{DD}}\right)^2} \qquad t_{PHL} = \frac{a_n\, C}{K_n\, V_{DD}}$$

$$a_p = \frac{2}{\frac{7}{4} - 3\left|\frac{V_{tp}}{V_{DD}}\right| + \left|\frac{V_{tp}}{V_{DD}}\right|^2} \qquad t_{PLH} = \frac{a_p\, C}{K_p\, V_{DD}} \qquad t_P = \frac{t_{PLH} + t_{PHL}}{2}$$

Method 2: *Equivalent Resistance*

$$t_{PHL} = 0.693 \times R_N \times C \qquad t_{PLH} = 0.693 \times R_P \times C \qquad t_P = \frac{t_{PLH} + t_{PHL}}{2}$$

$$R_N = \frac{R_{eff,N} \times \left(\frac{W_{eff}}{L_{eff}}\right)}{\left(\frac{W_n}{L_n}\right)} \qquad R_P = \frac{R_{eff,P} \times \left(\frac{W_{eff}}{L_{eff}}\right)}{\left(\frac{W_p}{L_p}\right)}$$

Capacitance Estimation: Inverter 1 feeding Inverter 2 with Wire

$$C_{int} = 2C_{gd,p1} + 2C_{gd,n1} + C_{dB,p1} + C_{dB,n1} \quad C_{ext} = \left(C_{wire} + C_{gs,p2} + C_{gs,n2}\right) \times \text{Fan_out}$$

$$C_{TOT} = C_{int} + C_{ext}$$

Transistor Sizing

Optimal Inverter

1. Choose L to be minimum length permitted by Fab
2. To strictly minimize inverter area while still fitting contacts: $1 < \left(\frac{W}{L}\right)_n < 2$
3. Trade area versus matching: $\left(\frac{W}{L}\right)_p \approx 2 \times \left(\frac{W}{L}\right)_n$
4. Increase $\left(\frac{W}{L}\right)_n$ to reduce R and therefore propagation delay
5. Then $n = \left(\frac{W}{L}\right)_{n,inv}$ and $p = \left(\frac{W}{L}\right)_{p,inv}$

Optimal Logic Circuit

1. Find longest series path through PUN = u; Reduce path resistance by this factor by increasing width of each W = u x p
2. Find longest series path through PDN = d; Reduce path resistance by this factor by increasing width of each W = d x n
3. If not a single-path device, if switch in parallel with "s" series devices, then make its resistance path the same -> $W = W_{ser}/s$

Transient Power Dissipation

$$P_{ave} = \frac{E_{diss,cycle}}{T_{clock}} = \frac{E_{diss,ON} + E_{diss,OFF}}{T_{clock}} = \frac{\frac{1}{2} \times C \times V_{DD}^2 + \frac{1}{2} \times C \times V_{DD}^2}{T_{clock}} = f_{clock} \times C \times V_{DD}^2$$